

Suppose $A = \{1,2\}$, $B = \{2,3\}$

b. $P(A) = P(\{1,2\}) = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$

d. $P(A \times B) = P(\{(1,2), (1,3), (2,2), (2,3)\})$

$$= \{\emptyset, \{(1,2)\}, \{(1,3)\}, \{(2,2)\}, \{(2,3)\}, \quad (1)$$

$$\{(1,2), (1,3)\}, \{(1,2), (2,2)\}, \{(1,2), (2,3)\}, \dots\} \quad (2)$$

you get the idea.

a. $P(\emptyset) = \{\emptyset\}$

b. $P(P(\emptyset)) = P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$

c. $P(P(P(\emptyset))) = P(\{\emptyset, \{\emptyset\}\}) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$

Let $A = \{a,b\}$, $B = \{1,2\}$, $C = \{2,3\}$ $A \times (B \cap C) = \{a,b\} \times \{2\} = \{(a,2), (b,2)\}$

$x = 1,3,5$, $y = a, b, c$, d $g: X \rightarrow Y$ defined by $1 \mapsto b$; $3 \mapsto b$; $5 \mapsto b$;

a. domain of g : $1, 3, 5$ co-domain of g :

b. $g(1) = b$ $g(2) = b$ $g(3) = b$

range of g : is 3 an inverse image of a ? Nah is 3 an inverse image of b ?

Yes

What is the inverse image of b ? $1,3,5$ What is the inverse image of c ? $\emptyset$

Represent g as a set of ordered pairs: $(1,b), (3,b), (5,b)$

Find all functions from $X = a,b,c$ to $Y = u$; ATTACH SKETCH

Find all the functions from $X = a,b,c$ to $Y = u, v$

Let F and G be the functions from $\mathbb{R}$ to $\mathbb{R}$ Define new functions $F - G: \mathbb{R} \rightarrow \mathbb{R}$ and $G - F: \mathbb{R} \rightarrow \mathbb{R}$ as: $\forall x \in \mathbb{R}, (F - G)(x) = F(x) - G(x)$
 $(G - F)(x) = G(x) - F(x)$

Does $F - G = G - F$ No.